Restaurant Fast-Food Prototype Scorecard Spreadsheet Review

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Summary of Major Comments

The following comments represent errors found when comparing values from the scorecard to the source cited in the scorecard:

- Wall criteria for 1978-1997 vintage should not be for offices (FF22).
- Infiltration I_75 value for 2008-2018 and 2019+ vintages inconsistent with cited source (FF25).
- Hot water consumption is much lower than source (FF55).

The following are the highest priority comments:

- Number of peak kitchen occupants is too low (FF15).
- Number of peak dining occupants is too low (FF16).
- Zoning is oversimplified (FF21).
- Total static pressure does not include typical adjustments (FF41).
- Hood exhaust rate too low for appliances (F48).
- Kitchen electric and gas use too low and should use bottom-up analysis (F60).
- Kitchen should operate longer hours than dining room for prep and cleaning time (F61).
- Infiltration schedule shows 0% and 100% when buildings are closed (F62).
- Exhaust schedule should not be zero when gas cooking schedule positive (F66).
- Lighting should be on when restaurant open, not modulating with occupancy (F64).

The following comments represent other important issues found:

- Window-to-wall ratio too low (FF11).
- Fast-food restaurants are not usually square buildings (FF6).
- Roofs are usually flat, not pitched (FF7).
- Wood-framed walls seem atypical (FF23).
- Drive-thru window infiltration should not be ignored (FF28).

- Kitchen exhaust summary number not consistent with calculations (F46).
- Number of parking spaces and related exterior lighting is too low (F57).

In addition, the following comments describe the lack of disclosed assumptions that make verifying the values in the scorecard nearly impossible:

- Please site sources for the major geometric assumptions (FF13)
- How is the foundation insulated (FF31)?
- What are the details for the make-up air units (FF39)?
- Is there transfer air from dining room into kitchen (FF43)?
- Is demand-controlled kitchen ventilation used (FF44)?

Background

The review of the "scorecard," which is a spreadsheet summarizing the model assumptions, is broken down by spreadsheet tab, which includes:

- Notes
- Prototype
- Zones
- Envelope
- HVAC System
- Thermostat
- Ventilation
- Interior Lighting
- Water Heater
- Ext Lighting
- Equipment
- Schedules
- Energy Usage
- Weights

The comments on these tabs are labeled using the following format FFx-<TAB>-<CELLRANGE>), where <TAB> is the name of the tab in the spreadsheet and <CELLRANGE> indicates the specific cells. Note that no tabs were reviewed after the Schedules tab.

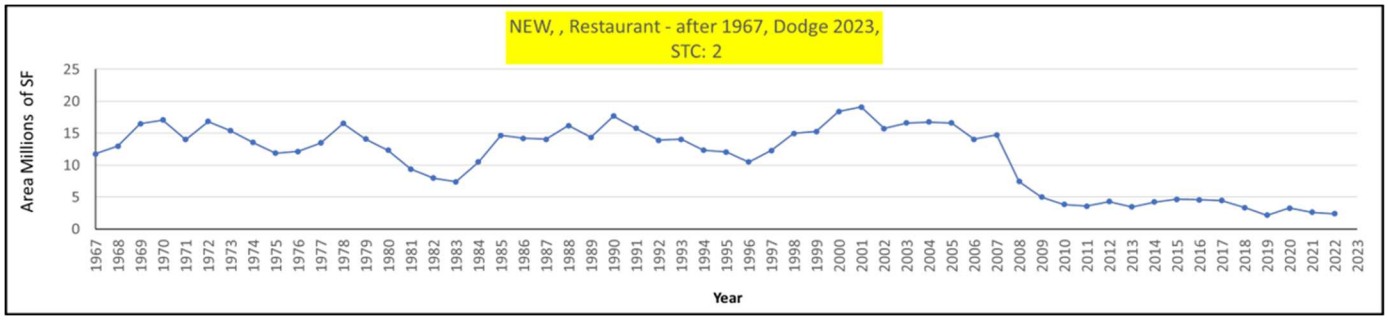
A copy of the spreadsheet is available that includes which cells on each tab were checked against the sources and were consistent (shown in light green) and which had anomalies and are described in a comment below (shown in tan).

FF1-Notes (B2:C5, C10:C11, B16:C16)

Several referenced sources do not include URLs.

FF2-Notes (B33:C42)

The descriptor "STC: 2" in the graph is not explained:



FF3-Notes (A46:C46)

The term "fast-food" is not typically used within the industry, which favors the term "quick service restaurant". This is implied in the notes included in cell B46: "Is a subcategory of food service building used for prepared-food sales. These places are often quick-service operations (e.g., drive-throughs, counter-service eateries,...)."

FF4-Notes (n/a)

Fast-food restaurants are a unique building type in that major chains often have standard plans used by franchisers to construct buildings. Obtaining some of these standard plans for the major fast-food chains would provide a set of assumptions on the prototype that would be quite valuable.

FF5-Notes (n/a)

To provide a thorough review of the prototype, I reviewed several great resources shown below prior to beginning my review. Originally, 85 resources were identified, but the following seemed the most important.

Sources that *should* be reviewed for background information and are fundamental for reviewing the school prototype are the following:

- Technical Support Document: 50% Energy Savings for Quick-Service Restaurants Zhang, et. al., 2010 (PNNL-19809) https://www.pnnl.gov/main/publications/external/technical_reports/pnnl-19809.pdf
- Assessment of Energy Consumption and Greenhouse Gas Emissions in a UK Quick-Service Restaurant Using EnergyPlus. Elias Eid, et.al. 2025 https://doi.org/10.3390/en18061377
- What's Cookin'? A Suite of Gas Efficiency Technologies for Commercial Kitchens, Mary Horsey, 20212 ACEEE Summer Study. https://www.aceee.org/wp-content/uploads/proceedings-1980-2020/2012/0193-000284.pdf
- ASHRAE 1362-RP report "Revised Heat Gain and Capture and Containment Exhaust Rates from Typical Commercial Cooking Appliances" (2008)
- ASHRAE 1469-RP Comfort in Commercial Kitchens (January 2013)
- ASHRAE 1631-RP Countertop Commercial Appliance Emissions - July 2016 - Jianshun Zheng, Kwanghoon Han, Bing Guo, Yan Zhang, Changfu Wu
- Energy Reduction in Commercial Kitchens, Denis Livchak, February 2017 https://frontierfstc.com/ceccook/Energy_Reduction_in_Commercial_Kitchens_SFIA.pdf
- Food Service Building Asset Rating Methodology and Analysis - Supriya Goel, Juan Gonzalez Matamoros, Alex Vlachokostas, Na Wang https://www.osti.gov/servlets/purl/1459906
- ASHRAE Handbook Application – Chapter 34 Kitchen Ventilation – 2023
- ASHRAE Handbook Refrigeration – Chapter 16 Food Service And General Commercial Refrigeration Equipment - 2026

Additional useful sources could be added to the list shown in the scorecard spreadsheet:

- Restaurant Energy Use Benchmarking Guideline, Roger Hedrick, 2011 https://docs.nlr.gov/docs/fy11osti/50547.pdf

- Title 24 Complex HVAC Energy Models and Resources, Neil Bulger, 2026 https://calbem.ibpsa.us/wp-content/uploads/2026/09/CBECC-Complex-HVAC-Example-Guide.pdf
- Selecting & Sizing Exhaust Hoods Improving Commercial Kitchen Ventilation (CKV) System Performance https://caenergywise.com/design-guides/ckv/CKV-Design-Guide-1_Selecting_Sizing_Hoods.pdf
- Commercial Kitchen Electrification Cost Study, December 18, 2025 https://www.smcsustainability.org/wp-content/uploads/Commercial-Kitchen-Electrification-Cost-Study.pdf
- Codes and Standards Enhancement (CASE) Initiative 2025 California Energy Code - Commercial Kitchens – Coakley. 2023. https://title24stakeholders.com/wp-content/uploads/2023/08/2025_T24_Final-CASE-Report-NR-Kitchens.pdf

FF6-Prototype (A10:C10)

The assumed aspect ratio of 1.0 (a square) does not match intuition or data from CBECS or PNNL's NC3 database presented in "Technical Support Document: 50% Energy Savings for Quick-Service Restaurants" (PNNL-19809), which supports an aspect ratio of 2 or 2.3.

According to the CBECS data, a rectangular footprint is very common design for QSRs (see Figure 2.2). The average building footprint aspect ratio of QSRs from the NC3 database is 2.3; and one of building floor plans is shown in Figure 2.3. The design experience of the authors who have worked with different QSRs also confirm that it is very common to design a QSR building with an aspect ratio of 2 and with roughly equal spaces for the dining and food-preparation areas. Therefore, rather than adopting the DOE Reference QSR design, we selected the prototype building floor plan shown in Figure 2.4.

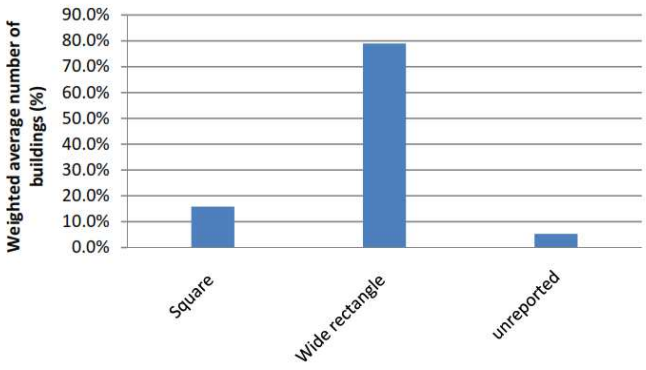


Figure 2.2. Building Shape Distribution of Post-1980 QSR Buildings (CBECS 2003)

The same report shows a zoning approach reflecting a 2:1 ratio.

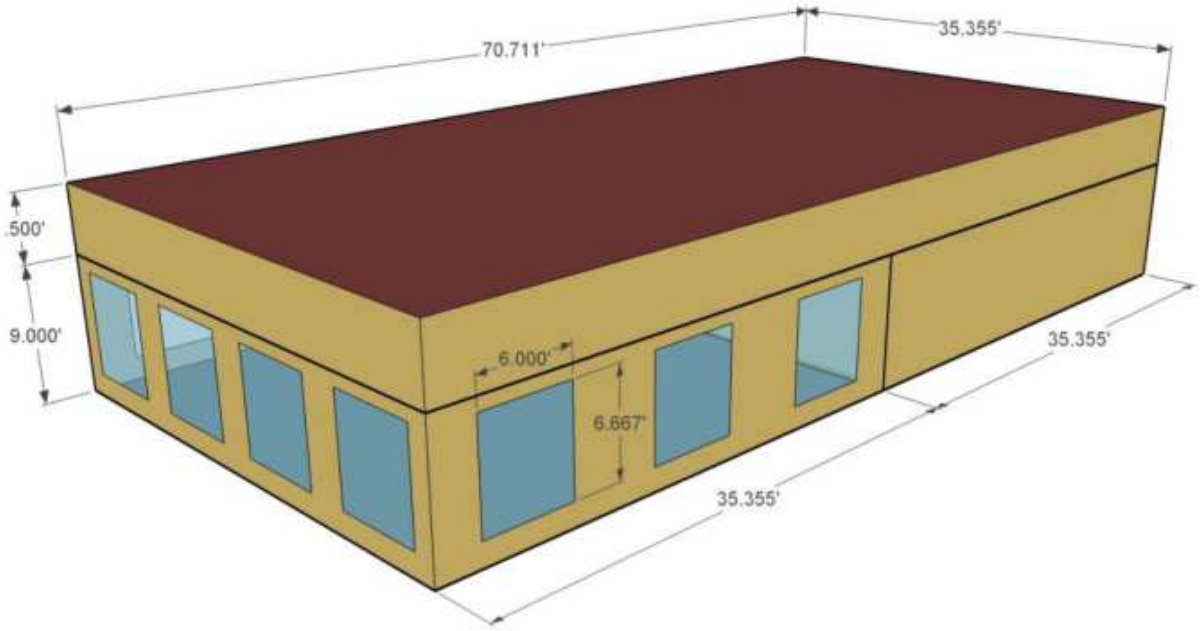
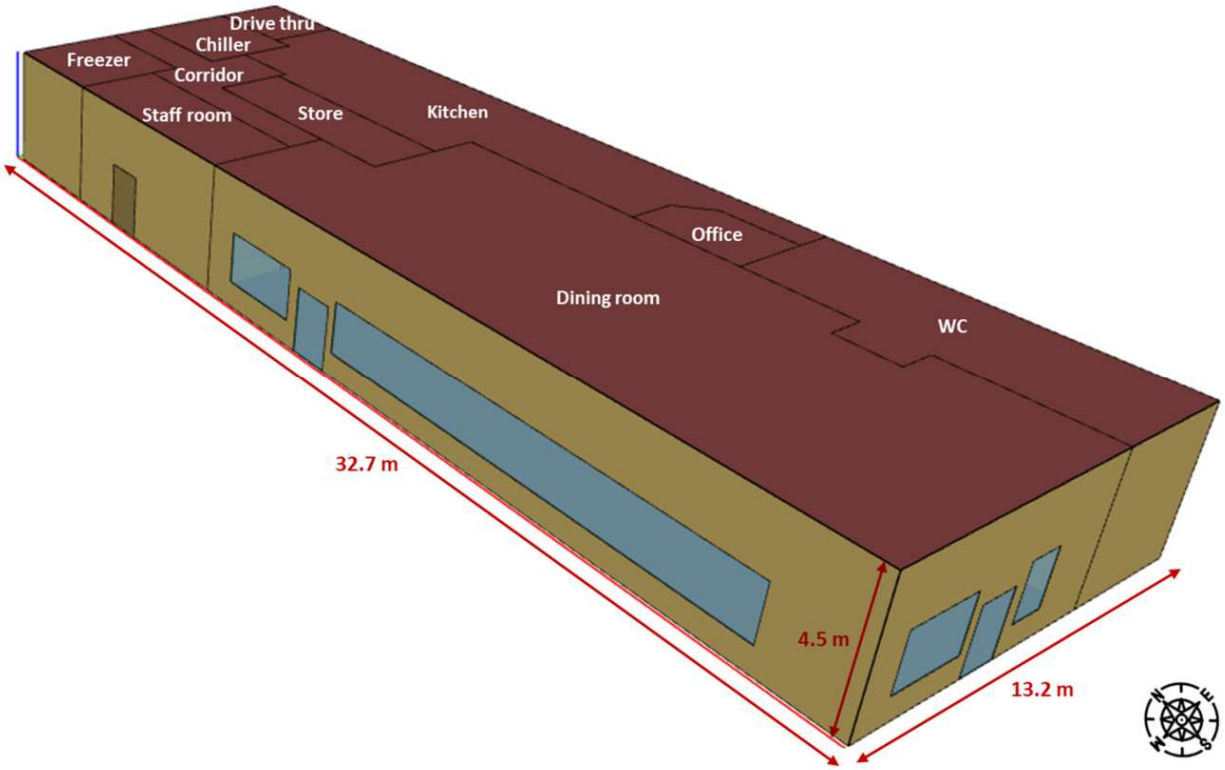


Figure 2.6. Axonometric View of the Prototype QSR Building

In "Assessment of Energy Consumption and Greenhouse Gas Emissions in a UK Quick-Service Restaurant Using EnergyPlus," the ratio is 2.5, as shown in the figure below from that paper:



In my personal experience, I can't remember ever being in a square fast-food restaurant; they are always long rectangles, and when they are not, it is noteworthy (usually an odd-shaped building on an odd-shaped lot). This may be partially due to the need for a drive-thru lane along one side of the building so that multiple cars can be between the ordering station and the pickup window.

FF7-Prototype (A12:C12, A29:B35)

The roof type shown is for an attic with a pitch of 17 degrees. This shallow-pitch roof is inconsistent with both personal memory of seeing many fast-food restaurants, as well as the data provided just a few rows below in the scorecard and repeated below. This prototype described by the scorecard is 2501 ft2. More than 50% of the fast-food restaurants that are 3500 sqft or less have flat roofs according to CBECS 2018, so it would be logical based on the data provided to assume that the roof is flat.

According to CBECS 2018, small restaurants mostly have flat or shallow pitch roofs.

CBECS 2018	Percentage of total weighted (Restaurant)		
Floor Area	Flat	Shallow pitch	Steeper pitch
<3500 sf	54%	14%	32%
>3500 sf	23%	15%	62%
All Restaurant	30%	15%	56%

In PNNL's "Technical Support Document: 50% Energy Savings for Quick-Service Restaurants" (PNNL-19809), they used the NC3 database, which, as a source, uses plans submitted to Dodge, and for all 11 quick-service restaurants in that source, they all had flat roofs.

FF8-Prototype (A38:E76)

Please add to the "Existing public prototypes" a comparison with the prototype described in "Technical Support Document: 50% Energy Savings for Quick-Service Restaurants" (PNNL-19809). This source provides another citable source for comparison or reference for almost all the assumptions needed in a prototype.

FF9-Prototype (n/a)

Please do not delete the data below row 27; it is useful background data for any reviewer of the scorecard.

FF10-Prototype (A38:E76)

Please add window-to-wall ratio and aspect ratio to the existing public prototype comparison table.

FF11-Prototype (A13:C13)

The window-to-wall ratio in the scorecard is 14% based on

- South: 28%
- East: 14%
- North: 0%

- West: 14%

Seems low when based on data from CBECS and shown in the PNNL "Technical Support Document: 50% Energy Savings for Quick-Service Restaurants" (PNNL-19809) shown below, which implies a higher average (although for that report, they also chose 14% WWR).

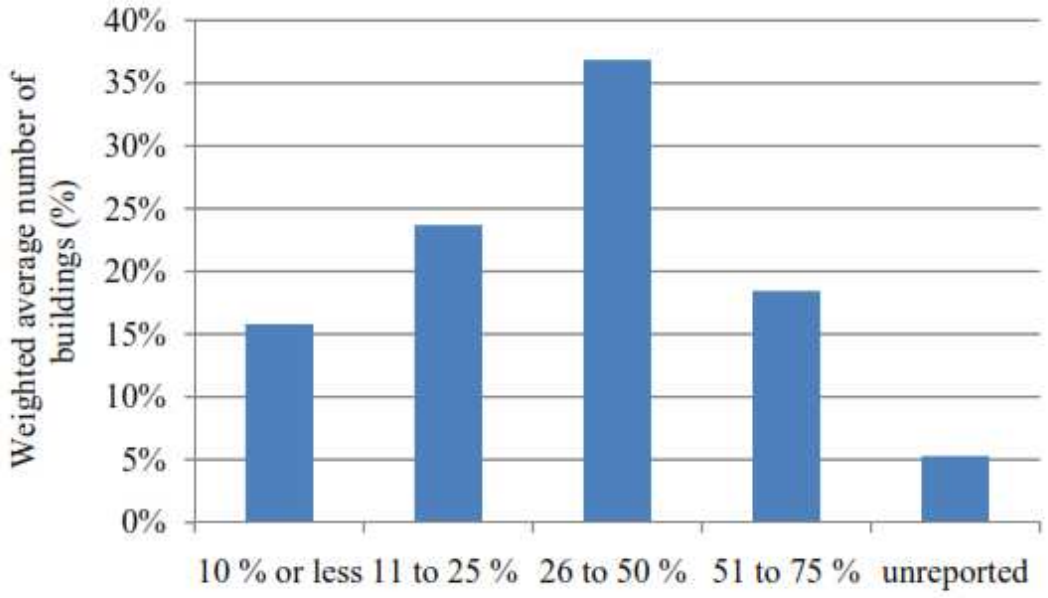


Figure 2.5. Window-to-Wall Area Fraction of Post-1980 QSR Buildings (CBECS 2003)

In general practice, the WWR is closely related to the exterior perimeter of the dining room as well as the table height (2.5 ft) and ceiling height (8ft), which is nearly all windows except where service (drink and garbage) and decorative areas. Based on this, from the interior perspective, most fast-food restaurants' dining rooms are probably 50% to 60% glazed.

FF12-Prototype (G2:G4)

The assumption of 50% dining and 50% kitchen should be substantiated; for many common fast-food restaurants, the kitchen zone, which usually includes the cashier counter area, is more than 50% of the overall area of the building. It would be good to provide a source for this assumption other than being consistent with other prototypes.

FF13-Prototype (A8:C17)

The shape, total floor area, aspect ratio, number of floors, roof type, window-to-wall ratio, shading geometry, floor-to-floor height, floor-to-ceiling height, and glazing sill height are shown as "Assumption; same as CEC Prototype 2022," which is not very useful or ultimately verifiable. Instead, it would be good to create the basis for what assumptions should be made for all of these based on original data, engineering judgment, or another source.

FF14-Prototype (n/a)

The type of fast-food restaurant is important for both the operating hours and for the kitchen load and exhaust rates. Almost all fast-food restaurants provide lunch and dinner, but many fast-food restaurants do not provide breakfast. Obviously, this directly impacts the number of operating hours for the building and all the cooking equipment. Whether or not the prototype serves breakfast should be disclosed on the tab.

In addition, a typical "hamburger and fries" type of fast-food restaurant is one of the most common but, from a kitchen perspective, also is one of the most intensive.

FF15-Zones (K5)

The scorecard shows 4 people in the kitchen of 1250 sqft based on Title 24-2026 Table 120.1A related to ventilation. This is a design number of people and should be the maximum. I do not think it is sufficient if it includes cashiers, drive-thru window workers, cooks, and management. I would estimate that most fast-food restaurants at peak capacity would have 3 cashiers, 1 drive-thru worker, 4 cooks, and a manager. This totals nine people. During most operations, half of this number is probably sufficient.

FF16-Zones (K3:L3)

The dining person density of 33 persons/1000 ft2 is equivalent to 30 sqft per person. According to this page:

https://www.timewornusa.com/6-steps-to-help-design-your-restaurant-seating-plan/

And the relevant portion of the page:

Step 2: Estimate restaurant seating capacity

Use people-per-square-foot guidelines to create a first pass range, then tune it for your concept and dwell time. As a rule of thumb, the more courseware and service you provide, the lower the density. The faster the turnover, the higher the density you can achieve, provided ADA access and safe server aisles remain intact.

- **Counter service / Fine dining:** about 1 person per **18 to 20 ft²**
- **Full service:** about 1 person per **15 to 18 ft²**
- **Commercial cafeteria:** about 1 person per **12 to 15 ft²**
- **Fast food / Banquet seating:** about 1 person per **10 to 11 ft²**

Using 11 ft2 /person results in 113 people, not the 41 shown in the scorecard. Even assuming that 25% of the dining room is space not included in the zoning plan (restrooms, vestibule, corridors, etc.) and therefore is not actually part of the dining space, and using 11 ft per person, that would result in 85 people, which is more than double the 41 people shown for the dining room in the scorecard.

FF17-Zones (A5:M5)

If the scorecard is changed to use a flat roof, then the attic row should be removed.

FF18-Zones (I5)

The attic volume computed, 25,000 ft3, assumes an average height of 10 ft for the entire 2500 sqft floor of the attic. For a low sloping roof of 17 degrees, the peak height would be 7.6 ft, and the average height would be 3.8 ft for a volume of 9554 ft3. The formula in that cell uses the floor-to-floor height on the prototype tab by mistake. If the aspect ratio is adjusted, this volume would be further reduced.

FF19-Zones (I6, A6)

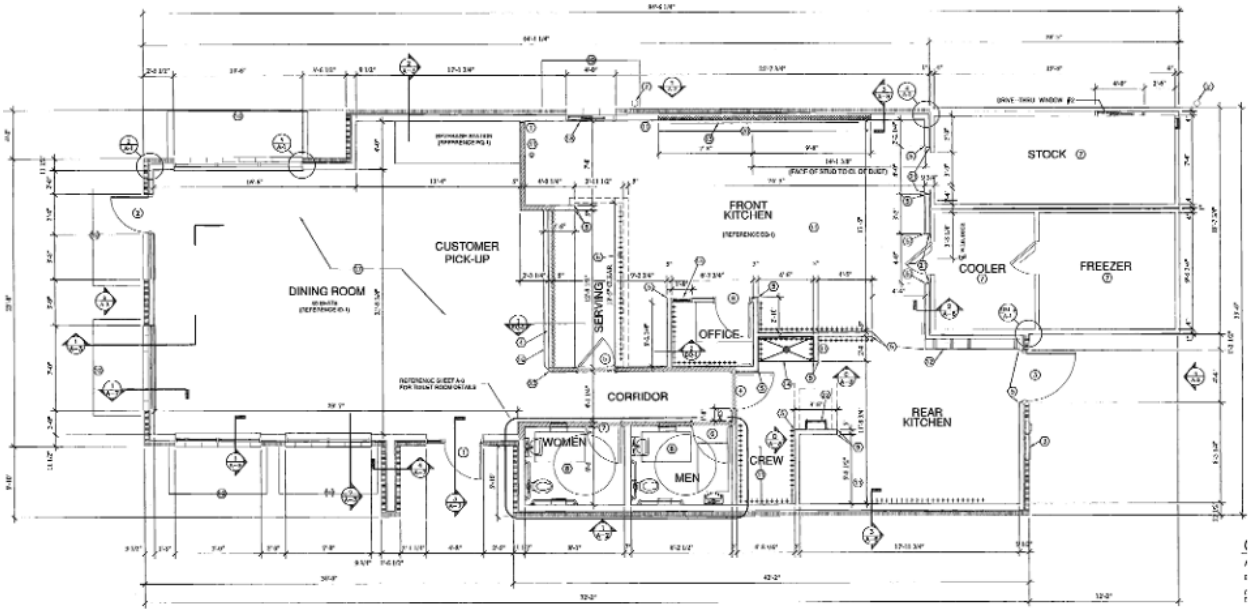
The volume of the building does not include the attic in the formula and ends up being the volume of the occupied zones only. That is fine if the row wasn't labeled "Total (3 Space)," which would include the attic.

FF20-Zones (C2)

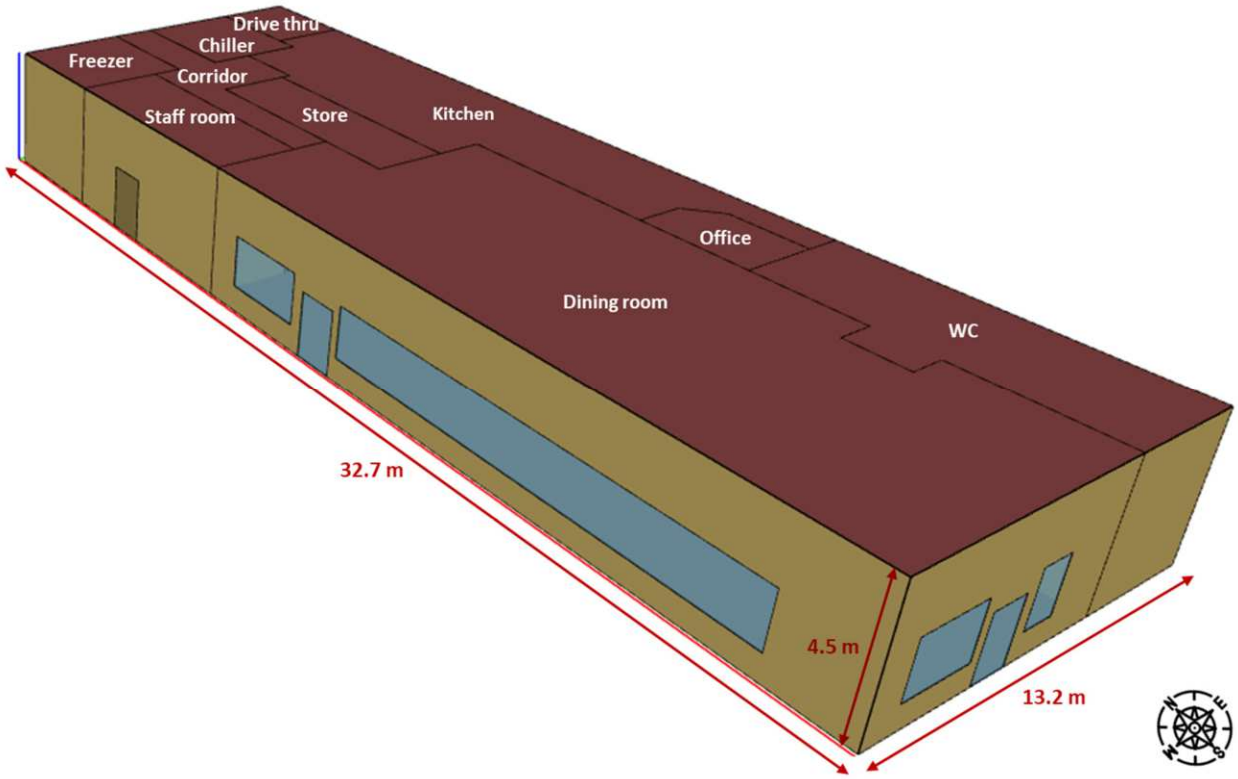
The cell reads "Space Type Primarily from [T24, 2022 Appendix 5.4A] - See Notes column," but there are no notes in the Notes column.

FF21-Zones (n/a)

Assuming just dining and kitchen, maybe oversimplifying the spaces in a fast-food restaurant, which should also include restrooms, stock rooms, cooler, freezer, office, and crew room (break room). This can be seen in the figure from "Technical Support Document: 50% Energy Savings for Quick-Service Restaurants" (PNNL-19809):



In "Assessment of Energy Consumption and Greenhouse Gas Emissions in a UK Quick-Service Restaurant Using EnergyPlus"



FF22-Envelope (H4:X4)

The exterior wall values for wood-framed walls for the 1978-1997 vintage using Title 24-1996 say that the values shown are "Copied from low-rise office alternative table: Package A with heat capacity of 0.0-3.99 (Btu/F-ft^2)," but the table that the criteria come from is specifically for office buildings, and restaurants are not covered. These are not the criteria from that vintage that would be appropriate for a restaurant.

FF23-Envelope (C4:D7)

The use of wood-framed walls seems unlikely for typical commercial building construction, which usually uses steel-framed walls.

FF24-Envelope (H8:X9)

The roof values are based on Title 24-1992, as explained in the note: "Considering 30 years for roof/ insulation replacement," and the values are correct, but the presumption that T24-1992 is a good stand-in for this in 2025 seems incorrect. If one assumes that all these roofs

were replaced 30 years ago, then Title 24-1995 would make more sense. Since the odds of all the roofs being replaced in 30 years are slight, a better assumption would be:

- The average value of roof requirements from Title 24 from 1995, 1998, 2001, 2005, 2008, 2010, 2013, 2016, 2019, 2022, or
- The value of roof requirements from 15 years ago, so using Title 24-2010 (less accurate but still justifiable).

This comment is the same as with other prototypes.

FF25-Envelope (G29:H30)

The I_75 values for infiltration for the 2008-2018 vintage and the 2019+ vintage are shown as 1.0 and 0.6, respectively. The sources cited for these are: "2022 CASE report (Table 38) & T24, 2008-2016 (Table 140.3-B)" and "2022 CASE report (Table 38) & T24, 2025 (Table 140.3-B)." Unfortunately, these values do not appear in the sources. The CASE report shows:

Table 38: Recommended Leakage Rates for Compliance Software.

#	Scenario	Recommended Infiltration Rate to be used in Compliance Software (cfm/ft² at 75Pa; infiltration through 6 sides)	Option to Select in Compliance Software
A	No continuous air barrier	1.1	Option 1
B	Continuous air barrier – no field verification or whole building leakage testing	0.7	Option 2
C	Continuous air barrier – field verification/inspection	0.5	Option 3
D	Continuous air barrier –whole-building leakage testing, default credit	0.4	Standard Design
E	Continuous air barrier – whole-building leakage testing, actual result	Tested result under 0.4	Option 4

And does not have a value of 1.0 or 0.6 in any option. The requirement for an air barrier in Title 24 varies from being required in some climate zones to all climate zones. If that is the case, the table on the Envelope tab of the scorecard should be climate zone specific.

FF26-Envelope (G25)

Typo in the word "sided."

FF27-Envelope (Z26:AA26, I26:I30)

The assumed roof area being the same as the floor area is not necessarily valid for a building with a pitched roof. The area of the roof is slightly larger, and this adds a 1.6% error to the I-design values shown in the scorecard. The six-sided pressure envelope for a building is generally considered the outer skin of the building. It is possible that the attic space is not considered part of the pressure boundary; if that is the case, the I-design values are correct, but this additional assumption should be documented.

FF28-Envelope (n/a)

Drive-through window (drive-thru) causes significant infiltration. These windows are generally open for much of the interaction with each car and, even when closed, are usually not a very good seal. They can be a very significant source of infiltration for a fast-food restaurant.

FF29-Envelope (D8)

The use of a pitched roof with an attic does not seem consistent with experience or the statistics presented on the Prototype tab (see FF7-Proto).

FF30-Envelope (n/a)

What solar reflectance is assumed for existing and new vintages of the building? This can significantly impact the cooling load on a building.

FF31-Envelope (n/a)

No description is provided for the foundation or any insulation required for the foundation.

FF32-HVAC System (G3:G4)

The use of PSZ with furnace and constant air volume for existing vintages makes sense, but please clarify which table in CEUS is being referenced when the source is stated as "CEUS

2022 + Engineering Judgment." It would be good to state what the engineering judgment is as well.

FF33-HVAC System (G5:G6)

The source of "T24, 2025" is very difficult to verify since it is probably referring to specific provisions or tables. Please specify the provisions.

FF34-HVAC System (H5)

This cell contains the words "Demand Control Ventilation" and is in the row for new construction dining room. There is no heading, so it is not clear if it is entered in error or if it belongs as part of the "other systems" portion of the table.

FF35-HVAC System (F3:F6)

The "other systems" column seems to provide details that are probably related to requirements. Please cite the sources, including the tables or section numbers.

HVAC system - RestaurantFastFood						
Vintage	**Space/Thermal Zone**	**Heating**	**Cooling**	**Air Distribution**	**Other Systems**	**Source**
Existing	Dining	Gas Furnace (PSZAC)	DX Coil (PSZAC)	CAV (PSZAC)	Economizer for after 2008	CEUS 2022 + Engineering Judgement
	Kitchen	Gas Furnace (PSZAC)	DX Coil (PSZAC)	CAV (PSZAC)	Makeup Air Unit (Combo with RTU) to supply make-up air exceeding transfer air from dining space (can be included in system13b)	
New Construction	Dining	Gas Furnace (PSZAC)	DX Coil (PSZAC)	CAV (PSZAC)	Economizer	T24, 2025
	Kitchen	Gas Furnace (PSZAC)	DX Coil (PSZAC)	CAV (PSZAC)	HR-Makeup Air Unit (systems that integrate make-up air and heat recovery functions) (can be included in system13b)	

The requirements related to the make-up air unit and a heat recovery make-up air unit are especially important to understand.

FF36-HVAC System (C13:C18)

The use of an 80% efficiency furnace is probably a reasonable assumption, but the source of that assumption should be a requirement and documented in the source column of the table. It is also not clear if this is combustion or thermal efficiency.

FF37-HVAC System (G14:G18)

The Notes shown in the column related to the economizer's max position and deductions for EERs with heating sections should have citations. In addition, perhaps a column specifically for economizer maximum position should be added.

FF38-HVAC System (F23)

The cell says "Fan Motor and Drive Efficiency [ACM 2025, Table 13]" but the rows below it are textual and do not appear to be related to ACM 2025.

FF39-HVAC System (n/a)

The make-up air units with (new construction) and without (existing vintage) heat recovery are not described anywhere. The exhaust and make-up air systems for kitchens can be quite complicated and deserve some specific assumptions. This is especially the case since a large fraction of the total energy use of a restaurant is related to kitchen ventilation.

FF40-HVAC System (C24:C25, C28:C29)

The systems are described as "CAV Kitchen (system <7,437 cfm)" and "CAV Dining (system <7,437 cfm)" but no justification is provided for the airflow being less than 7437 cfm.

FF41-HVAC System (D24, D28)

The total static pressure of 2.5 is a generic value from the PNNL 20405 report for any system that is less than 7437 cfm and constant volume. For fast-food restaurants with more complicated ventilation systems, this should be derived more carefully with assumptions that are specific to the exhaust and ventilation requirements. Table 17 of the ACM describes "Adjustment Credits, All Other Fan Systems," which includes exhaust filters, scrubbers, or other exhaust treatment and energy recovery, which may both be applied.

FF42-Thermostat (A7:S10)

Due to the high temperatures of the equipment and the high ventilation rates, the actual heating and cooling setpoints in kitchens can sometimes not be all that relevant. The actual conditions in the kitchen are often quite warm, and the occupants often expect that

to be the case. The suggested heating and cooling setpoints of 66F and 79F acknowledge this to a point, but it would not be surprising if data can be found that would make the cooling setpoint even higher.

It is not clear if high kitchen temperatures in summer months mean that the setpoint is normal and the cooling capacity is just insufficient, or if the setpoint is raised since the cooling system is not helping anyway. One thing is clear is that temperatures are often very high in kitchens. ASHRAE 1469-RP "Comfort in Commercial Kitchens" (January 2013) showed that the average air temperature in quick service (fast-food) restaurants was even higher than the proposed setpoint.

Table 7: Average of Physical Parameters by Kitchen Type and Thermal Zone in Summer

Summer Kitchens		t_o		t_a		t_{mr}		RH	v_a		I_{cl}	activity
Kitchen Type	**Kitchen Zone**	[°C]	[°F]	[°C]	[°F]	[°C]	[°F]	[%]	[m/s]	[ft/min]	[clo]	[met]
Casual	Cooking	34.9	94.8	32.7	90.8	39.4	102.9	40	0.42	83	0.6	3.9
	Preparation	28.7	83.7	28.5	83.3	29.2	84.6	43	0.53	104	0.7	2.8
	Dishwashing	28.7	83.6	29.0	84.3	28.2	82.8	71	0.14	28	0.6	2.0
Institutional	Cooking	30.9	87.7	28.6	83.5	34.3	93.7	36	0.39	77	0.7	3.8
	Preparation	24.0	75.2	23.6	74.5	24.4	76.0	42	0.30	59	0.7	2.9
	Dishwashing	24.7	76.5	24.2	75.6	25.3	77.5	52	0.35	69	0.6	3.2
QSR	Cooking	29.1	84.3	27.7	81.9	31.1	88.0	45	0.31	61	0.6	3.1
	Preparation	26.5	79.8	26.1	78.9	27.0	80.6	43	0.23	45	0.6	2.5
	Dishwashing	21.4	70.6	21.5	70.8	21.4	70.4	51	0.18	35	0.7	2.7

FF43-Ventilation (n/a)

It is important to model the transfer of air between the dining room and kitchen since that is the most common configuration. Usually, the dining room is pressurized slightly positive compared to the kitchen.

FF44-Ventilation (n/a)

Demand-controlled kitchen ventilation is commonly used in many kitchens to modulate the amount of ventilation. It is different than typical demand control since it is based on the use of cooking operations (surface or duct temperatures) but can have a significant impact on energy use.

FF45-Ventilation (F3, I3)

The description "'2 restroom, each 25/50 cfm pr unit" is referencing private rather than public restrooms. The value actually used in cell I3 is 50 cfm, which is correct.

Table 120.1-B – Minimum Exhaust Rates [ASHRAE 62.1: Table 6-2]

Occupancy Category	Exhaust Rate, cfm/unit	Exhaust Rate, cfm/ft²	Air Class	Notes
Toilets – private	25/50	-	2	E
Toilets – public	50/70	-	2	D

D – Rate is per water closet, urinal, or both. Provide the higher rate where periods of heavy use are expected to occur. The lower rate shall be permitted to be used otherwise.

E – Rate is for a toilet room intended to be occupied by one person at a time. For continuous systems operation during hours of use, the lower rate shall be permitted to be used. Otherwise the higher rate shall be used.

FF46-Ventilation (I4)

The "Total cfm Driven by Exhaust" for the kitchen is shown as 875 cfm, but just a few rows below that, the hood exhaust rate (I12) shows 3203 cfm. It really should be the higher number.

FF47-Ventilation (n/a)

It is very likely that a make-up air unit is part of the exhaust air design of a fast-food kitchen. Please describe how that would be modeled. It is likely that it would include a heat recovery system for new construction (see HVAC system tab).

FF48-Ventilation (C12)

While the 210 max air flow rate (cfm/ft) is correct from T24, 2025, Table 140.9-C, the value might be lower than typical. The ASHRAE 1362-RP report "Revised Heat Gain and Capture and Containment Exhaust Rates from Typical Commercial Cooking Appliances" (2008) shows:

Appliances classified as light duty and requiring 200 cfm/ft of exhaust air to ventilate the cooking process include:

- ovens, electric and gas, (including bake, roasting, revolving rack, rethermalizing, convection, combination, counter-top conveyor-baking/finishing, and deck)
- braising pans/tilting skillets, electric and gas
- dishwashers, rack conveyor
- ranges, electric element (with or without oven), induction cook tops
- rotisseries, electric and gas
- rice cookers, electric and gas
- rethermalizers, electric and gas
- cheesemelters, electric and gas
- salamanders, electric and gas
- steamers, electric and gas compartment (both pressure and atmospheric types)
- steam-jacketed kettles, tabletop (20 gallon and less)

Appliances classified as medium duty and requiring 300 cfm/ft of exhaust air to ventilate the cooking process include:

- griddles, electric and gas
- double-sided griddles, electric and gas
- fryers, electric and gas (including open deep-fat, donut, kettle, and pressure)
- pasta cookers, electric and gas
- chain (conveyor) broilers, electric and gas
- underfired broilers, electric
- dishwasher, door type
- range, gas open-burner (with or without oven)
- hot-top ranges, electric and gas
- smoker, electric, pressurized
- steam-jacketed kettle, floor mounted (over 20 gallon), electric and gas

Appliances classified as heavy duty and requiring 400 cfm/ft of exhaust air to ventilate the cooking process include:

- underfired broilers, gas
- wok ranges, electric and gas
- overfired (upright) broilers, electric and gas
- conveyor pizza ovens, electric and gas

Where most of the appliances under the hood are in the 300 or 400 cfm/ft range, this same list also appears in the ASHRAE Handbook chapter on Kitchen Ventilation (Applications Chapter 34).

FF49-Interior Lights (G4)

For existing vintages, the assumed source was Title 24-2016, which implies that buildings older than 10 years have been updated to that version of the standard. The likelihood of the dining area of a fast-food restaurant being updated every 10 years is a fair assumption. It is probably more likely for large chain restaurants, including those franchised, to update their lighting to make sure the restaurant remains appealing to customers. For non-chain, independent fast-food restaurants, this assumption is probably less valid. Those independent fast-food restaurants, in my experience, may look considerably older and are often less concerned with the aesthetics of the dining area than they are with providing food. I would suggest that further investigation be made into both the split of independent and chain fast-food restaurants (based on some Google searches, about 30% to 40% of fast-food is independent in California) and how often updates are made to independent fast-food restaurant dining rooms.

FF50-Interior Lights (G5)

The kitchen lighting for existing buildings (all vintages) uses Title 24-2016, and that is probably not a reasonable assumption. Updating the lighting system in a kitchen is unlikely to be a priority for investment for any restaurant. It would not be surprising for fast-food kitchens to have the original or nearly original lighting system since improvements to this system cost money but do not increase customers (like updating the lighting system in the dining room might) or functionality (like near kitchen appliances). In addition, this is an important assumption since the graph of restaurant construction by year that is shown on the Notes tab implies that many fast-food restaurants were built earlier than 2007. A better assumption would be to either use a significantly older version of Title 24 or use a weighted average for a range of years.

FF51-Interior Lights (n/a)

Simplifying the model to just dining and kitchen probably has a significant impact on the overall computed lighting power density of the building. A fast-food restaurant includes many other space types such as storage, break room, restrooms, corridor, entrance, office, coolers, etc.

FF52-Interior Lights (C3, F3)

The heading of the column for new construction and existing vintages of "Additional LPD for Portable Lighting (W/ft^2)" should probably say "Additional Lighting Power Allowance" since the additional allowances are not related to them being portable.

FF53-Water Heater (n/a)

While lots of information is shown, the connection between tables is not always clear.

FF54-Water Heater (G20:H38)

The table shown for "Sizing a Tank-Type Water Heater for Various Restaurants" is cited as being from "Design Guide: Advanced Water Heating for Foodservice, 2022, Table 8," but that reference is not included on the Notes tab, and no link is provided. I did find a copy here:

https://web.archive.org/web/20240715100546/https://www.caenergywise.com/design-guides/Advanced-Water-Heating-Design-Guide.pdf

A newer version from 2023 is referenced right next to the 2022 version, so it is probably not necessary. The 2023 version is included in the Notes.

FF55-Water Heater (E9:E11)

The gal/hr per person shows:

	Peak Hc
Space Name	**Hot Water Usage Specification (gal/hr per person) [T24, 2022, Appendix 5.4A , sheet: "2022 SpaceFuncData-Input"]**
Dining	0.12
Kitchen	N/A

But the values for both should be 0.578 gas/hr per person according to the citation. This doesn't impact the kitchen gas usage since that is derived from a different table, but the number of people in the dining room pushes the peak water usage slightly above 75 gal/hr to 77.45 gal/hr.

FF56-Water Heater (F10:F11)

The gas storage water peak usage of 58.6 gallons/hour (4.8 dining, 53.6 kitchen) is smaller than typical according to "Technical Support Document: 50% Energy Savings for Quick-Service Restaurants" (PNNL-19809), which used 82 gallons/hour:

> Average hot water consumption varies significantly among individual QSRs, ranging from 250 to 1200 gallon per day (Delagah and Fisher 2010). Based on field monitoring, the average "burger-based" QSR is estimated to use 500 gallon per day (FSTC 2007). The peak hourly hot water demand is 82 gal/h (5.17 L/s). This is based on field-monitored data using a typical hot water load profile on an average weekday (see Figure 3.12).

FF57-Ext Lights (B11:C11)

Providing adequate parking for customers and employees is critical for a successful restaurant, including fast-food. One rule of thumb is 1 parking space per 4 seats plus 1 per employee would be 14 parking spaces based on the occupant density from the Zones tab. Assuming the 85 people seated (see FF16-Zone), that would be 25 parking spaces. According to this source:

https://graphingparking.com/2013/02/06/parking-requirements-for-restaurants/

The median is 20 spaces based on parking requirements across the country, with several California cities being 30 or more spaces (this is for a 2500 ft2 restaurant with 90 seats).

The "Technical Support Document: 50% Energy Savings for Quick-Service Restaurants" (PNNL-19809) showed 19,553 ft2 for parking.

The scorecard assumes 10 parking spaces. Assuming four of those spaces are for employees (the number shown for kitchen occupants), that would mean the 6 remaining spaces would be used to fill the 41 seats, or 7 people arriving per car, which seems unrealistic.

FF58-Ext Lights (B12:C12)

The assumption of one building entrance in a fast-food restaurant does not seem practical. Many fast-food restaurants have two doors for customers to enter the facility (either on both sides of the building or two on one side. In addition, employees almost always have a separate entrance toward the rear of the building, often toward the garbage containers. I

would recommend that three entrances be assumed. It is often necessary to provide multiple exits based on fire codes.

FF59-Equipment (E4)

The gas power density of 35.08 is double the value from the referenced "T24 2025 - Appendix 5.4A: "2022 SpaceFuncData-Input." The cell formula shows the ratio of the total building divided by the area of the kitchen. It is not clear why this adjustment is being made. The original 17.54 Btu/hr-ft2 shown in the referenced source is an energy intensity and is intended to be multiplied by the floor area of the kitchen.

FF60-Equipment (n/a)

A bottom-up analysis of the equipment (cooking, dishwasher, appliance, refrigeration) would be much more justifiable than a value intended to represent an average commercial kitchen. The ventilation tab already has a list of appliances that could be used as a start. It is likely that a walk-in chiller and walk-in freezer are present in most fast-food restaurants.

Several great sources that could be used for this include:

- Technical Support Document: 50% Energy Savings for Quick-Service Restaurants" (PNNL-19809)
- What's Cookin'? A Suite of Gas Efficiency Technologies for Commercial Kitchens - Mary Horsey (ACEEE 2012)
- ASHRAE Research Project 1362 Revised Heat Gain and Capture and Containment Exhaust Rates from Typical Commercial Cooking Appliances – Fisher and Swierczyna, 2008
- ASHRAE Research Project Report 1631-RP Countertop Commercial Appliance Emissions - July 2016 - Jianshun Zheng, Kwanghoon Han, Bing Guo, Yan Zhang, Changfu Wu
- Energy Reduction in Commercial Kitchens - Denis Livchak - 2017
- Food Service Building Asset Rating Methodology and Analysis - Supriya Goel, Juan Gonzalez Matamoros, Alex Vlachokostas, Na Wang
- ASHRAE Handbook Application – Chapter 34 Kitchen Ventilation – 2023
- ASHRAE Handbook Refrigeration – Chapter 16 Food Service And General Commercial Refrigeration Equipment – 2026

The bottom-up analysis will probably show that the plug load power densities and the gas power density for the kitchen are much too low. Several of these references include total electric and gas densities, and they are significantly higher than what is shown on this tab.

FF61-Schedules (A4:AC6, A32:AC34)

The kitchen and dining room occupancy schedules should differ. The dining room has little or no occupancy prior to the opening time as well as after closing time, but kitchen staff need to prep and close down (including cleaning) the kitchen. In my experience, kitchens generally open one to two hours before opening time and one hour after closing time. This will have cascading effects on other schedules based on occupancy.

FF62-Schedules (A16:AC18, A47:AC49)

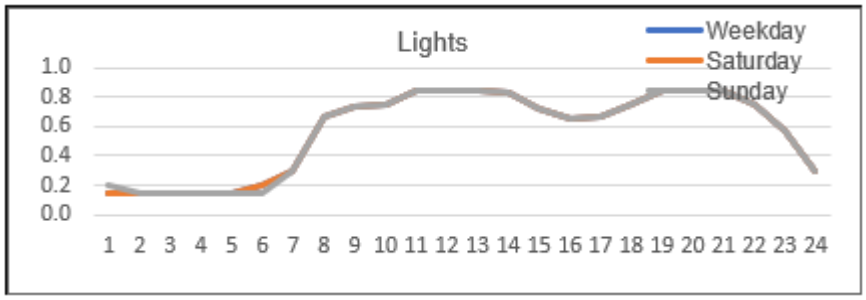
The infiltration schedule seems odd: the fan system is off for several hours overnight, but the infiltration is either zero or one during those hours. It should probably be one for any hour the fan system is off. But pressurization differences between the dining room (slightly positive) and the kitchen (slightly negative) should mean that the kitchen infiltration is always one when the fan system (and exhaust hoods) are on.

FF63-Schedules (R4, X4:Z4, Y5)

Peak occupancy of the dining room at 80% every weekday for an hour at lunch and three hours for dinner and 90% for one hour each Saturday at dinner seems optimistic to me. Certainly, fast-food restaurants can be quite busy, but that many hours at 80% or higher seems optimistic. Also, I think many fast-food restaurants are busiest for lunch rather than dinner.

FF64-Schedules (K7:AC9, K35:AC37)

The lighting schedule for both the dining room and the kitchen changes during the occupied hours. Unless this is intended to represent some daylighting strategy, it seems unlikely to occur in a small fast-food restaurant. Normally, I would expect the lights in the dining room and kitchen to turn on when occupied (slightly earlier and later for the kitchen than the dining room) and off (to only egress lighting) for non-occupied hours.



The "Technical Support Document: 50% Energy Savings for Quick-Service Restaurants" (PNNL-19809) shows the following lighting schedule:

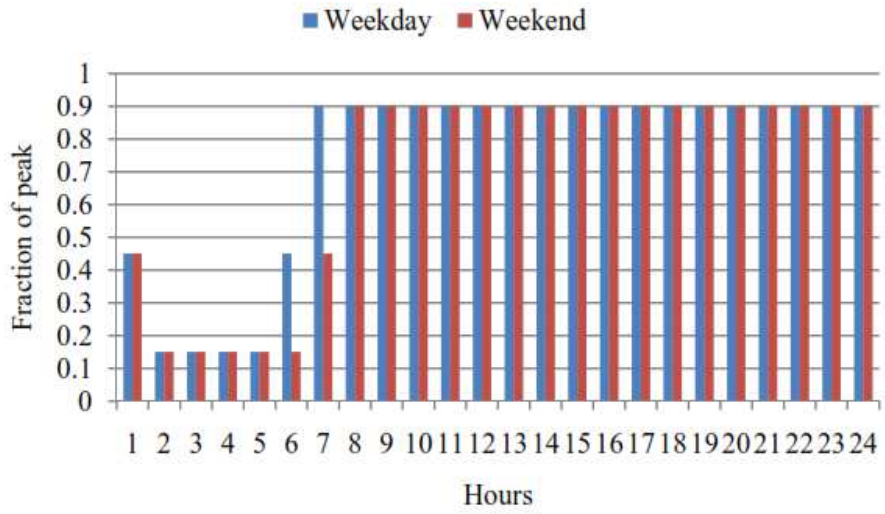


Figure 3.1. Typical Lighting Schedule for Baseline

FF65-Schedules (n/a)

If a bottom-up analysis is being used for kitchen equipment, some schedules have been developed in "Technical Support Document: 50% Energy Savings for Quick-Service Restaurants" (PNNL-19809) that show profiles by equipment:

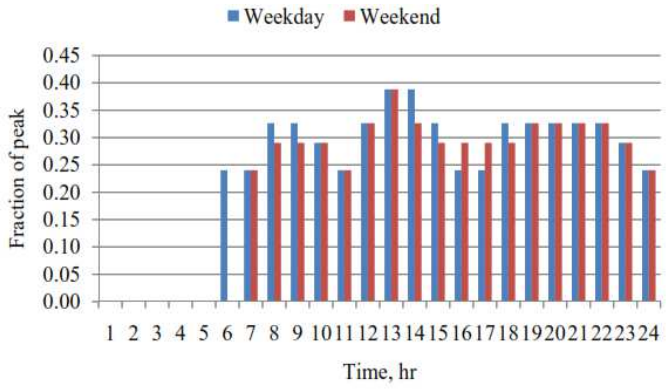


Figure 3.5. Baseline Fractional Usage Profile for the Hooded Cooking Griddle

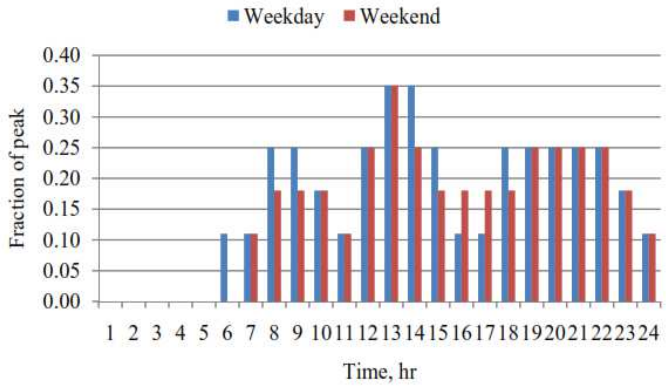


Figure 3.6. Baseline Fractional Usage Profile for the Hooded Cooking Fryer

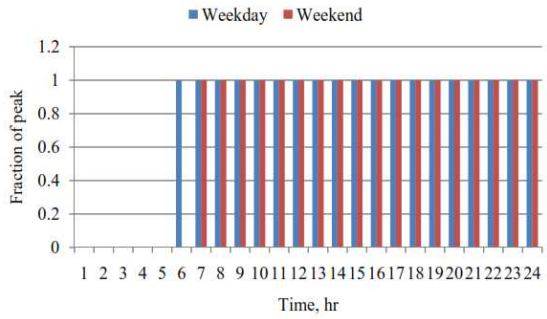


Figure 3.7. Baseline Fractional Usage Profile for the Unhooded Cooking Equipment that Operates Whil the Restaurant is Open

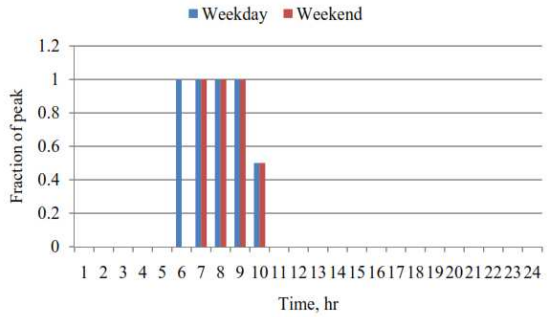


Figure 3.8. Baseline Fractional Usage Profile for the Unhooded Cooking Equipment only Used During Breakfast Hours

FF66-Schedules (Y50:AC55)

The ExhaustHoodEqualOrLessThan5000cfm schedule for the operation of the hood is zero (off) in the evening while gas cooking is at 50-90%.